

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester MTech. (CSE)(CBCS) Degree Examinations
 January 2023

22CSE102 – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is the Physical Symbol System? What is the AI technique?	10	L*1	1	1
b)	How does AI exploit knowledge? Explain in detail the turing Test.	10	L1	1	1
2. a)	What are the main aspects considered before solving a complex AI problem? What is state space representation in AI?	10	L2	1	1
b)	Identify a heuristic function for the traveling salesman problem.	10	L2	1	1
Unit – II					
3. a)	Explain that DFS is neither complete nor optimal search.	10	L1	2	1,2
b)	What is a Production System? Describe different classes of Production Systems with examples.	10	L1	2	1,2
4. a)	List out the types of AI Agents. Explain any two with suitable examples and neat diagram.	10	L1	2	1,2
b)	What is a uniform cost search? Differentiate between Uninformed and Informed Search techniques.	10	L1	2	1,2
Unit – III					
5. a)	How is AI related to Knowledge? Differentiate between Declarative and Procedural Knowledge.	10	L2	3	1,2
b)	Explain the rules of Intelligent Agents and PEAS Representation with examples.	10	L1	3	1,2
6. a)	Explain Water Jug Problem using a state space search. Generate Production rules for this problem.	10	L2	3	2
b)	What's the trade-off between bias and variance? Explain how a ROC curve works.	10	L2	3	2
Unit – IV					
7. a)	Why do we need data preprocessing? Explain Data Preprocessing in Machine learning.	10	L2	4	2
b)	Discuss the difference between Linear Regression and Logistic Regression.	10	L2	4	2
8. a)	Discuss the challenges in Machine Learning. Explain different perspectives and issues in machine learning.	10	L2	4	1
b)	Explain the steps in designing a learning system.	10	L2	4	1
Unit - V					
9. a)	Explain the two key difficulties that arise while estimating the Accuracy of Hypothesis.	10	L2	5	2
b)	How does the decision tree algorithm work? Explain.	10	L2	5	1
10. a)	Discuss the Naïve Bayes algorithm with an example.	10	L2	5	2
b)	Explain the derivation of the K-means algorithm.	10	L2	5	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
